

TechNova Data Analytics Internship

Task 03 – Financial KPI Dashboard

1. Project Overview

This project was completed as **Task 03 of the TechNova Data Analytics Internship Program**. The objective was to analyze a financial business dataset, define key performance indicators (KPIs), and develop visual dashboards to understand overall business performance.

The project uses the **Financial Sample dataset** and includes data cleaning, KPI calculation, visualization, and dashboard development using Python and Plotly Dash. An additional Power BI dashboard was also developed for interactive financial reporting.

2. Dataset

The dataset contains financial information related to products, countries, customer segments, sales, costs, discounts, and profit.

Dataset: *Financial Sample.xlsx*

After preprocessing:

- **Rows:** 700
- **Columns:** 16
- **Missing values:** 0
- **Duplicate rows:** 0

Main fields include Sales, Profit, COGS, Units Sold, Product, Country, Segment, Date, and Discount Band.

3. Data Preprocessing

The following preprocessing steps were performed using Python and Pandas:

- Loaded the Excel dataset.
- Cleaned column names.
- Converted the Date column into datetime format.
- Handled missing values in Discount Band.

- Checked for duplicate records.
- Prepared date-related fields for analysis.
- Calculated the required KPIs.

4. Key Performance Indicators

Five KPIs were selected to measure financial performance:

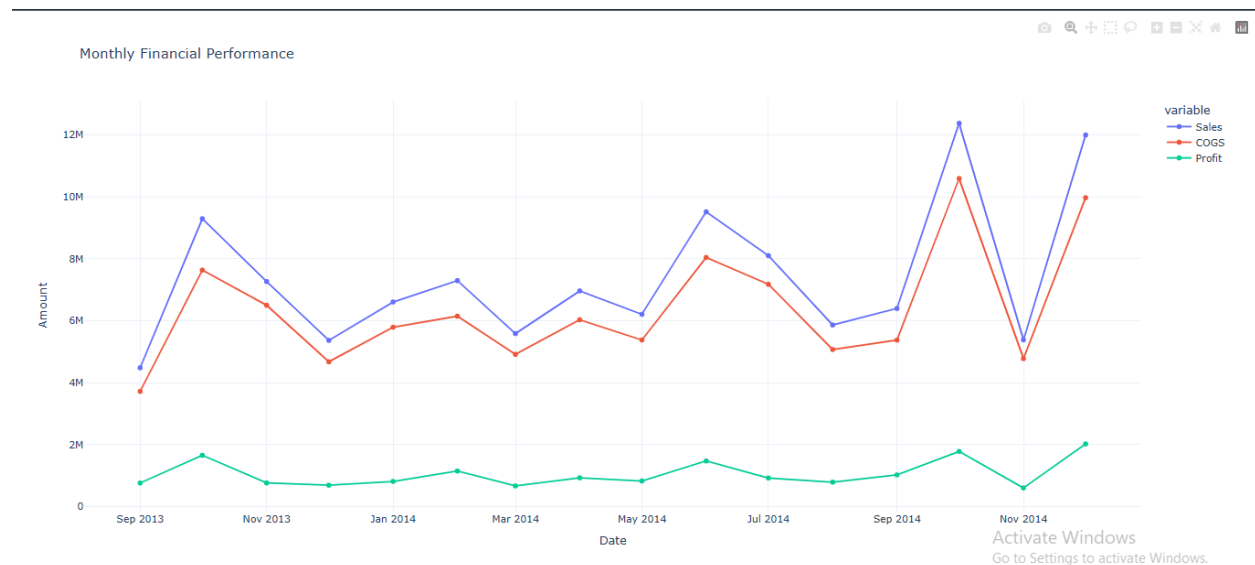
KPI	Result
Total Sales	\$118,726,350.26
Total Profit	\$16,893,702.26
Total COGS	\$101,832,648.00
Profit Margin	14.23%
Total Units Sold	1,125,806

5. Dashboard Development

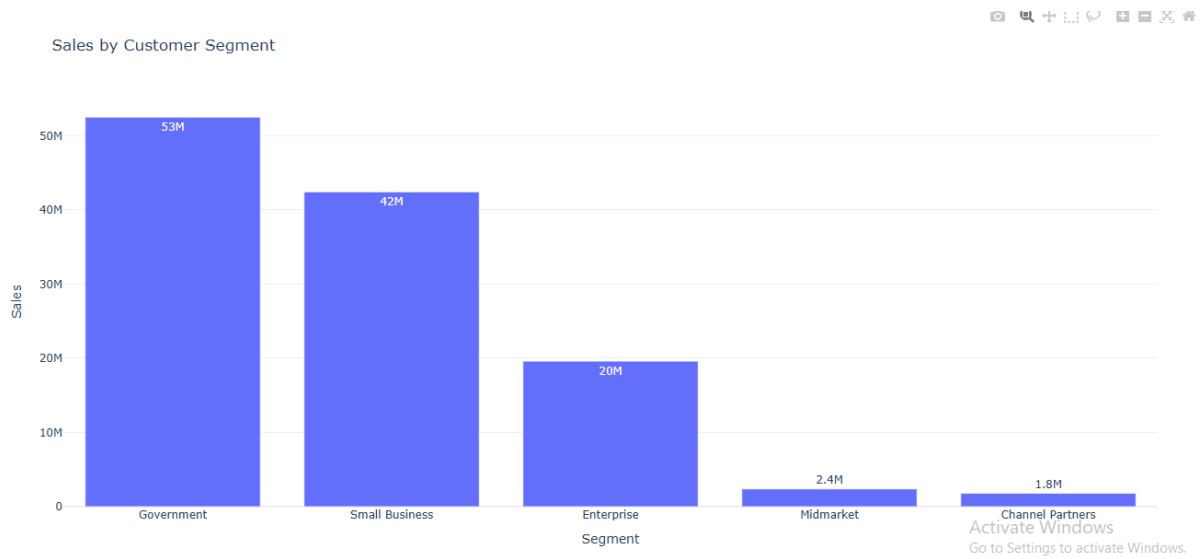
Plotly Dash

An interactive dashboard was developed using **Plotly Dash**. It provides:

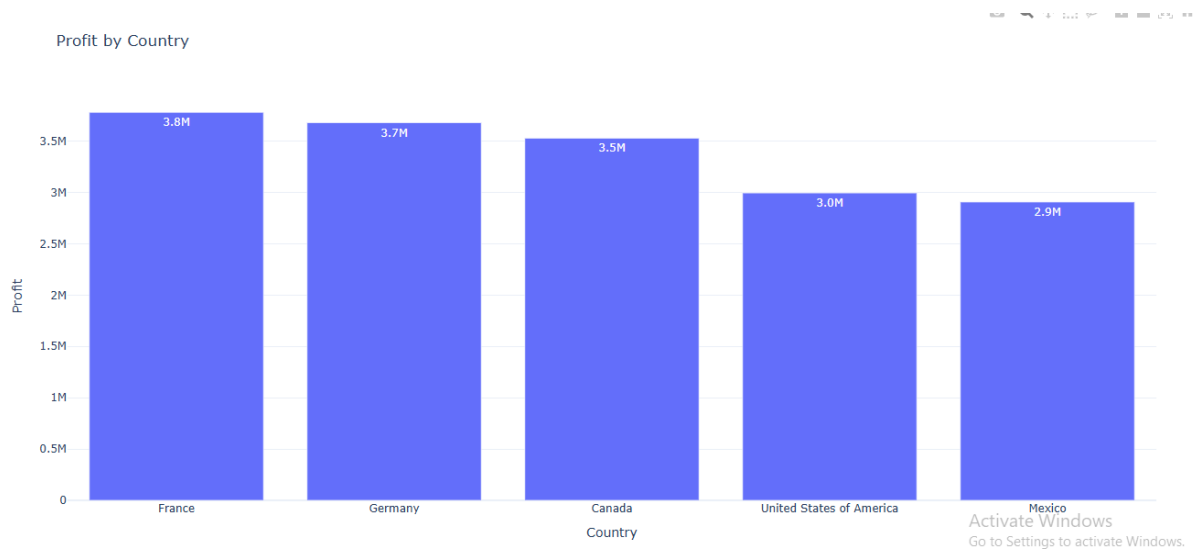
- KPI summary cards
- Monthly Sales, COGS and Profit trends



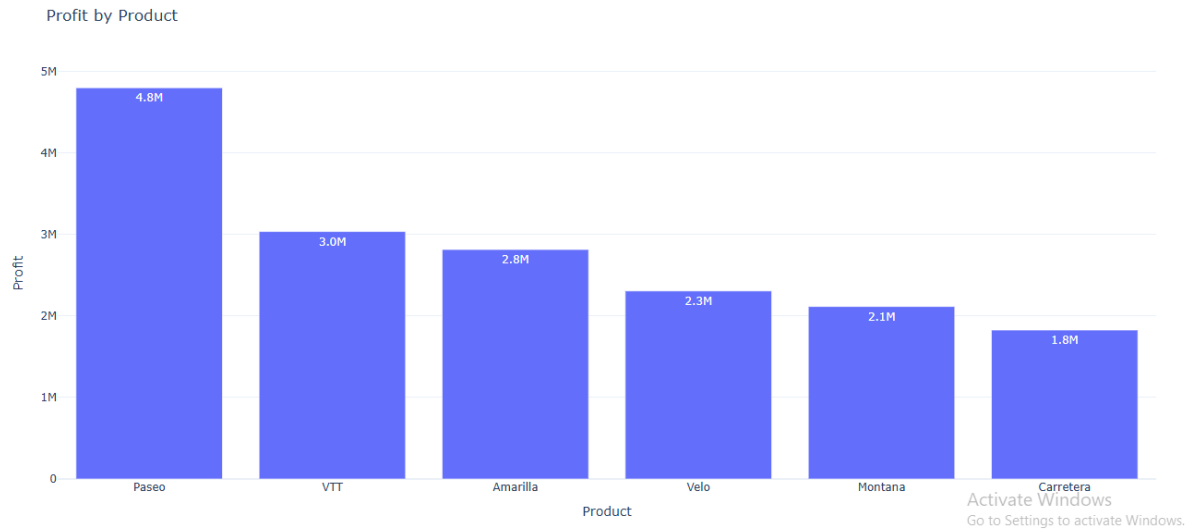
- Sales by customer segment



- Profit by country



- Profit by product



- Country, Segment, Product and Year filters

Power BI

An additional **Power BI dashboard** was created using the same financial dataset. It provides interactive KPI cards, slicers, financial trends, and comparative visualizations.

The .pbix dashboard file is included in the GitHub repository.

6. Tools & Technologies

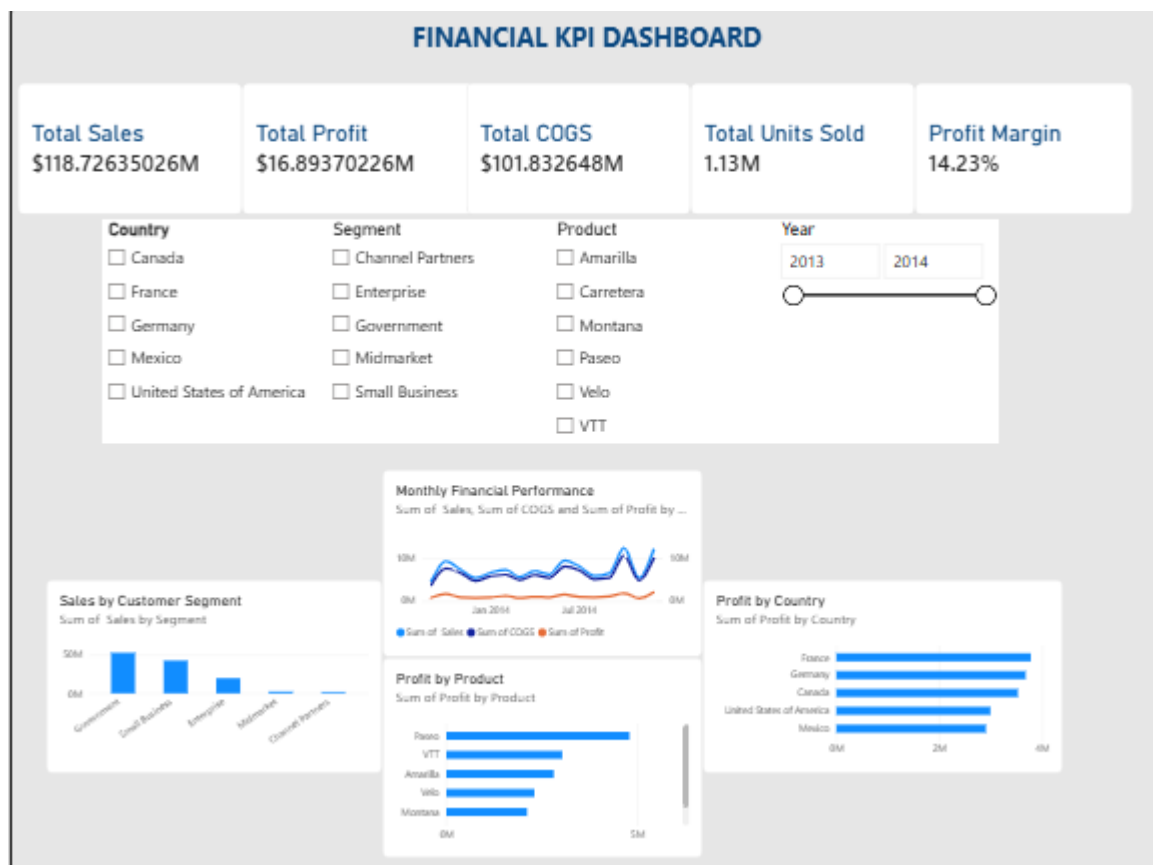
- Python
- Pandas
- NumPy
- Matplotlib
- Plotly
- Plotly Dash
- Power BI
- Microsoft Excel
- Colab Notebook

7. Key Outcomes

The project provided practical experience in:

- Financial data analysis
- KPI definition and calculation
- Data cleaning and preprocessing
- Interactive data visualization
- Dashboard development
- Business-oriented data presentation
- Using both **Plotly Dash** and **Power BI** for reporting

The completed dashboards provide an interactive way to explore financial performance across different products, countries, segments, and time periods.



8. Conclusion

The Financial KPI Dashboard successfully demonstrates the complete workflow from **raw financial data to business insights and interactive dashboards**. The project strengthened practical skills in Python-based analytics, KPI development, Plotly Dash, and Power BI.